Cover Letter: Submission to Applied Mathematics and Computation

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Dear Editor-in-Chief,

We are pleased to submit our manuscript entitled "LUNA: An Astronomy-Grounded Metaheuristic Optimization Algorithm Based on Real Lunar Orbital Dynamics" for consideration for publication in *Applied Mathematics and Computation*.

Summary of Contributions

This paper introduces LUNA (Lunar-inspired Update and Navigation Algorithm), a population-based metaheuristic whose search operators are mathematically derived from real lunar orbital mechanics. The vis-viva equation, Kepler's second law, lunar libration, and tidal syzygy are embedded *directly* into the update equations; in the original formulation, setting any astronomical variable to zero eliminates the corresponding astronomical modulation, leaving only residual Gaussian perturbation. This addresses the sustained critique by Sörensen (2015), Camacho-Villalón (2018), and subsequent metaphor-novelty frameworks that many "novel" nature-inspired algorithms are mathematical rebrandings of classical operators.

Theoretical Contributions

The paper establishes seven formal results with full proofs:

1. **Theorem 1** (Monotone Best-Fitness Decrease) and **Corollary 1** (Convergence under boundedness).
2. **Theorem 2** (Bounded Expected Exploration Radius) quantifying the libration perturbation's second moment.
3. **Theorem 3** (Exponential Step-Size Decay) proving that the vis-viva velocity decays as $\mathcal{O}(e^{-\delta t/(2T)})$.
4. **Theorem 5** (Finite Expected Hitting Time) providing the non-asymptotic bound $\mathbb{E}[\tau_\varepsilon] \leq R(1 + 1/p_\varepsilon^{\text{eff}})$.
5. **Theorem 6** (Weak Ergodicity) using the Dobrushin coefficient to establish that the LUNA Markov chain loses dependence on initial conditions.
6. **Theorem 7** (Almost-Sure Convergence to the Global Optimum) combining the hitting-time bound with a Borel–Cantelli argument.
7. **Corollary 2** (Polynomial Rate under Strong Convexity) showing that the expected hitting time scales as ε^{-D}, consistent with the curse of dimensionality.

To the best of our knowledge, among astronomy-inspired metaheuristics, these are the first explicit hitting-time and Markov-chain ergodicity guarantees. We explicitly acknowledge that analogous results for evolution strategies and simulated annealing have long been established in the broader stochastic-optimization literature, and we do not claim novelty relative to that literature.

A dedicated subsection (Section 5.6) provides five explicit theory-to-experiment connections, verifying that each theorem's prediction is qualitatively consistent with the corresponding empirical measurement.

Empirical Contributions

- **CEC 2022 ($D = 10$, 20 runs, 12 functions):** LUNA achieves Friedman rank #1 (1.83), outperforming DE (2.08), GA (2.50), and 7 other baselines; 11/12 wins against AVOA ($p < 0.05$, Wilcoxon).
- **High-dimensional experiments ($D = 50$ and $D = 100$):** LUNA remains competitive (Friedman ranks 5.17 and 5.83), defeating PSO, GSA, and SMA on the majority of functions. The dimensional degradation is consistent with the ε^{-D} rate predicted by Corollary 2.
- **IEEE CEC 2025 BC-SOP track ($D = 30$, 29 functions, U-score protocol):** LUNA again achieves rank #1 (Friedman 1.76, U-score -1058), demonstrating that its performance generalizes across three CEC benchmark generations (2017, 2022, 2025).
- **Statistical rigor:** Wilcoxon signed-rank, Friedman test ($p < 10^{-11}$), Holm–Bonferroni post-hoc, Nemenyi critical-difference diagram, and Dolan–Moré ECDF analysis (LUNA has the highest AUC, 998.94).
- **Effect size:** Cliff's delta for the LUNA-vs-DE comparison is -0.376 (medium), with a heterogeneous per-function split indicating that LUNA specializes on multi-modal functions while DE specializes on unimodal ones.
- **Ablation study:** 7 variants $\times$ 20 runs $\times$ 12 functions = 1680 trials; LUNA-full significantly outperforms LUNA-no-lateDE ($p < 1.7 \times 10^{-5}$) and LUNA-no-astronomy ($p < 1.9 \times 10^{-6}$, Holm-corrected).
- **Empirical hitting-time measurement:** $P(\tau_\varepsilon \leq t)$ curves across six ε thresholds verify Theorem 5's prediction qualitatively.

Fit with Applied Mathematics and Computation

The manuscript engages directly with the recent AMC literature: it cites 12 papers published in *Applied Mathematics and Computation* within the last five years (2020–2025), including the multi-strategy sine cosine algorithm (Chen et al., 2020), the jellyfish optimizer (Chou & Truong, 2021), accelerated DIRECT for constrained global optimization (Stripinis et al., 2021), consensus-based optimization (Bae et al., 2022), and recent convergence analyses of regularized Newton methods (Yamakawa & Yamashita, 2025). The theoretical framework draws on Markov-chain methodology that has appeared recently in AMC for molecular dynamics (Vandecasteele & Samaey, 2024) and risk measures (D'Amico & De Blasis, 2024).

Declaration

This manuscript has not been published elsewhere and is not under consideration by any other journal. All authors have approved the manuscript and agree with submission to *Applied Mathematics and Computation*. The code and benchmark data are publicly available at https://github.com/lutfananas/LUNA-Optimization-Algorithm to support reproducibility.

We thank the editor and reviewers for their time and consideration. We look forward to your response.

Sincerely,
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